

UNDERGRADUATE RESEARCH EXPERIENCE PROGRAM (UREP)

UREP Proposal #	UREP 32-0226-250233			
Project title in English	Signaling Use: A Study on Turn Signal Usage and Its Impact on Road Safety			
Project title in Arabic	دراسة سلوك واستخدام السائقين لإشارات الانعطاف وتأثيرها على السلامة المرورية			
Type of Application	☐ New ☒ Resubmission			
Type of Research	☐ Basic Research ☒ Applied Research ☐ Experimental development / Translational Research			
Submitting Institution	Qatar University, College of Engineering, Qatar Transportation and Traffic Safety Center			
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Research Plan

1. Summary

Road traffic crashes are a major global health challenge. The World Health Organization reported 1.19 million road traffic deaths in 2021, a 5% drop from 2010. Despite the decrease in road traffic deaths, it remains the leading killer of children and youth aged 5 to 29 years and globally 4-wheel vehicle occupants represent 30% of fatalities; followed by pedestrians who make up 23% of fatalities [1]. In Qatar, only in 2023, a total of 183 deaths, 494 major, and 8,745 minor casualties were reported from road traffic crashes [2]. Driver negligence and lack of attention, resulted in around 45% of crashes, and insufficient distance leading to rear-end collisions, responsible for around 20% of crashes were the primary cause of crashes [3].

Road safety is a systemic engineering problem comprising drivers, vehicles, roads, road furniture, driving environment, and traffic management. Drivers ensure safety by following traffic rules, such as obeying speed limits and using turn signals. However, failure to follow these rules affects not only individual drivers but also traffic as a whole. Turn signals as a means of communication among vehicles and with other road users are most widely used at intersections controlled by traffic lights, stop or yield signs, roundabouts, lane change, merging or diverging, and temporary stops. Turn signals alert others to a driver's intentions, giving them time to adjust. Studies revealed that failure to use turn signals while turning or lane changing can cause crashes which can be prevented [4], [5].

Among few studies carried out to investigate the use of turn signals at roundabouts, Belz et al., [6] studied the use at three single-lane roundabouts in the USA. They found that exiting vehicles, particularly those that do not use turn signals when departing from the circulating stream of traffic influence the entry decision of drivers on the same approach [6]. In Qatar, only 23% and 18% of drivers used signals when entering and exiting two-lane roundabouts respectively, with younger drivers (<25 years) signaling less frequently [7]. This researcher conducted another study on the factors affecting the use of turn signals and identified turn type, roundabout type, correct lane use, vehicle types, age, type of clothes, and distraction as a main factor affecting turn signal usage at modern roundabouts [8].

An observational study performed at intersections by collecting dynamic data in a naturalistic setting in the USA showed that 75% and 52% of drivers used turn signals while turning and lane changing respectively [5]. Similarly, research undertaken in Canada revealed, an overall rate of turn signal usage of 76% with values ranging from 54% to 95%. The usage was found lower for right-turning vehicles, heavy traffic volume, with the presence of a dedicated turning lane, and when following a vehicle not using a turn signal [9]. In Vietnam, turn signal use was higher for drivers (68%) than motorcyclists (40%) [10]. Further, drivers' beliefs, perceived risks, and environmental factors influenced turn signal use, but lifestyle behaviors, like smoking, did not [11]. The use of turn signals while merging lanes is also important. A study found that merging decisions are based on both objective factors (distance, speed) and subjective impressions (e.g., aggression) [12].

Most studies focused on limited scenarios for turn signal use. They fail to consider all areas where we can use turn signals to obtain a better understanding of turn signal usage patterns and their implications for road safety. No studies examined turn signal use in temporary stop areas. In Qatar, the study by [7], [8], only focused on turn signal usage at two-lane roundabouts. These studies used limited small sample sizes, and they did not consider important socio-demographic factors such as drivers' experience, and the age was approximated by the data collectors standing on the roadside while conducting an observational study. Traffic conditions and drivers' risk awareness were not considered.

2. Objectives/Significance

In this study, we aim to investigate two key aspects of turn signal usage at different road facilities and their implications for road safety. This may include intersections (signalized, unsignalized, and roundabout), lane changes, on-ramp, off-ramp, and temporary stops. First, we will analyze the signal use of a virtual car approaching the driver. We will also examine how this influences the driver's reactions, utilizing safety measurements such as Post-encroachment Time (PET), which is the time interval between when one vehicle enters the conflict zone minus the time the second vehicle enters the conflict zone, and Time-to-Collision (TTC), which is the estimated time remaining before a collision occurs if both vehicles continue on their current paths. Second, we will examine the actual use of turn signals by drivers at various locations, including

roundabouts and intersections, to understand the patterns of signal usage and correlation with socio-demographic factors in different traffic scenarios. This comprehensive approach will enhance our understanding of how turn signal behavior affects road safety and inform future policies and driver training programs.

3. Methodology

This project is proposed to be conducted using a driving simulator (hereinafter, “DS”) installed at the Qatar Transportation and Traffic Safety Center (QTTSC) at Qatar University. The DS enables researchers to study the behavior and interaction among road users and other roadway facilities by replicating various aspects of real-world driving conditions, including driving behavior, road geometry, traffic scenarios, weather conditions, and vehicle dynamics. DS was chosen to be used because it allows researchers to have a controlled environment, easy to collect a wide range of data, and accessibility. While conducting the experiment, participants will drive on the selected built roadway facilities, and the rate of turn signal use will be measured. Additionally, pre and post-experiment questionnaires will be administered to collect the subjective information. Figure 1 shows a sample of images where the drivers were supposed to use the turn signal.



a) Signalized intersection



b) Pull out on-street parking



c) Lane changing

Figure 1. Sample of images where drivers neglect the use of turn signals

a. Apparatus

Figure 2 shows the advanced DS in QTTSC, Qatar University which will be used to carry out the study. The simulator was validated by [13], [14] and used in various studies [13]; [14]; [15], [16]. The DS consists of a driving unit and three large screens (5760 × 1080 pixels). The screens have a refresh rate of 60 hertz with a 135° field of view (FOV) and are operated using SCANeR studio software. The driving unit is a fixed base type and is identical to a Range Rover Evoque, with a force feedback steering wheel, indicators, speedometer, pedals, and an automatic gearbox. The DS is capable of collecting 67 different parameters such as speed, headways lateral/longitudinal positions and acceleration, breaking characteristics. The DS is also incorporated with eye tracking and E4 wristband (for psychological measurements).



Figure 2. The advanced driving simulator at QTTSC, Qatar University

b. Scenario Design

This project classifies scenarios by facility type, analyzing driver signal usage at intersections and the impact of virtual vehicle signals during lane changes, merging, and temporary stops. To enhance the validity of our findings, other vehicles in the simulator will be programmed to exhibit a range of realistic driving behaviors, including critical events such as sudden lane changes, unexpected stops, and varied signaling approaches. These scenarios will be designed using SCANeR studio. Table 1 shows the study scenarios.

Table 1: Study scenarios

Facility types	Traffic control system	Driver turn signal usage scenarios	Facility types	Virtual vehicle turn signal usage scenarios
Intersections	Signalized	- Right turn (RT) - Left turn (LT)	Road links	- Lane changing (Overtaking)
	Unsignalized (stop or yield sign)	- Left turn (LT) - Right turn (RT) - U-turn (UT)		- Off-ramp: Diverging from traffic - On-ramp Merging into traffic

	Roundabout	- LT (Entry and Exit) - THR (Entry and Exit)	Temporary stops	- Pull into on-street parking - Pull out on-street parking
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c. Procedure

The ethical approval from the Institutional Review Board at Qatar University (QU-IRB) will be obtained for recruiting participants. Obtaining the Institutional Review Board (IRB) approval for our project will be managed by the research mentors of the project. We will target a minimum of 60 drivers to participate in the experiment. We aim to recruit a diverse participant pool through official invitation e-mails, targeting individuals aged 18 to 72 years, with a balanced representation across different age groups and genders to ensure the findings are representative of Qatar's driving population. Each participant will be tested individually with the following procedures:

1. Intake interview: After welcoming the participant, a brief introduction about the DS will be given by the research team. Then, the participant will be asked to fill out a pre-experiment questionnaire that focuses on socio-demographic background, driving experience, and knowledge of traffic rules. After that, the participant will be asked to sign an informed consent form which contains information about simulation sickness and the right of the participants to stop the experiment at any time. Moreover, the form gives the right to use the collected data for research purposes.
2. Driving simulator familiarization: The participant will be asked to undertake a 10-minute familiarization drive. This drive's main purpose is to familiarize participants with the DS controls before starting the experimental drives.
3. Experimental drives: Each participant will drive on a designed roadway facility of 15 km to 20 km in length. Scenarios under each road facility type will be designed in a networked roadway system. All participants will be instructed to drive according to normal traffic rules as they do in the real world.
4. Follow-up questionnaires: The participant will fill out A computer-based post-experiment questionnaire to collect further information about the DS and the individual's subjective evaluation.

d. Data collection and analysis

After each participant finishes the experiment, the raw data will be saved with specific names. Then, the required data for analysis purposes will be extracted into Excel sheets. The following data sets are expected to be collected for the analysis process:

- ✓ Questionnaires: This includes pre- and post-experiment questionnaires. In the pre-experiment questionnaire data related to socio-demographic background, driving experience, and knowledge of traffic rules will be collected. The post-experiment questionnaire includes subjective information on the driver's risk awareness of the use of turn signals.
- ✓ Number of turn signals used on each type of road facility by each participant. This will help us to determine the percentage of usage of turn signals and its correlation with socio-demographics.
- ✓ In addition to measuring turn signal usage, we will enhance our data collection to include traffic safety measures such as instances of hard braking, near-crashes, PET, TTC, and crashes, utilizing the simulator's capabilities to provide a comprehensive analysis of driver behavior and its impact on road safety.
- ✓ Turn signal timing and location: The distance the driver starts to indicate the turn signal from the intersection. These are collected for analysis of the use of turn signal behavior of the drivers considering surrounding traffic conditions.
- ✓ Since every participant will be tested for each condition and situation on the driving simulator, different statistical tests by IBM SPSS 28.0 and Minitab will be used for the analyses. Potential statistical models may include regression analysis to examine the relationship between turn signal usage and socio-demographic factors, as well as ANOVA and chi-square tests to compare differences across various driving scenarios.

To verify the results obtained in the simulator environment, we plan for a follow-up study that compares simulator findings with real-world driving data, acknowledging that this may require additional time and resources beyond the current project timeline.

e. Technical description by work packages

WP1 “Literature review”

Objectives: Review the recent literature about the use of turn signals and its impact on road safety.

Task 1.1 “Global and local studies conducted on the rate of turn signals usage and its impact on road safety”: In this task, road safety problems related to the usage of turn signals will be reviewed.

Task 1.2 “Implication of turn signal timing and location”: In this task, relevant studies will be reviewed.

WP2 “Design and coding of scenarios”

Objectives: To learn and design all the scenarios using SCANeR studio.

Task 2.1 “Scenario design”: This task includes obtaining the real-world roadway facilities videos to replicate and design them as a virtual environment. The obtained data will include road design (Highway and intersection geometry, temporary stopes, etc.), roadside elements (buildings, lights, etc.), and default signs from the Ministry of Transport (MOT) related to the use of turn signals.

Task 2.2 “Scenario coding”: Using SCANeR studio to design the scenarios used in this project.

Task 2.3: “Questionnaire development”: Developing Pre- and post-experiment questionnaires.

WP3 “Experiment Execution”

Objectives: To register the participants and to carry out the experiments

Task 3.1 “Participants recruitment”: Participants from the Qatar University community (students, faculty, and staff) and outside the university will be invited through official invitation e-mails.

Task 3.2 “Data collection”: In this task, data will be collected from the participants. This includes the data collected from the simulator and pre- and post-experiment questionnaires.

WP4 “Data analysis and reporting”

Objectives: To extract the collected data, analyse it, and report the results.

Task 4.1 “Data extraction”: In this task, the students will extract the required data from the primary raw data collected by the driving simulation software i.e., SCANeR studio.

Task 4.2 “Data analysis”: Students will use SPSS software to analyse the extracted data. They will be guided and supported about the suitable analysis tests for this project.

Task 4.3 “Reporting”: The results of this project will be submitted to a scientific paper for publication.

4. Role of the Undergraduate Researchers

This project will engage students from the departments of Civil and Environmental Engineering and Computer Science and Engineering. Each comprises three individuals. These six students have outstanding academic performance as shown below. They are ranked among the top 5% of undergraduate students in the departments.

- ✓ S1 (Mr. Rufus John Kurian), Computer Science & Engineering Department (GPA= 3.83/4)
- ✓ S2 (Mr. Majd Jeahd Alhakim), Computer Science & Engineering Department (GPA= 3.65/4)
- ✓ S3 (Mr. Mohammad Kasif), Computer Science & Engineering Department (GPA= 3.5/4)
- ✓ S4 (Mr. Osama Adel Mahmoud Mohamed), Civil & Environmental Engineering Department (GPA= 3.77/4)
- ✓ S5 (Mr. Yousef Ahmed), Civil & Environmental Engineering Department (GPA= 2.96/4)
- ✓ S6 (Mr. Kareem Darwish), Civil & Environmental Engineering Department (GPA= 2.95/4)

Different tasks will be assigned to the three students under the guidance and supervision of the mentor. We will implement a structured schedule of regular bi-weekly meetings where students can present their progress, discuss challenges they are facing, and receive constructive feedback from mentors and peers, ensuring continuous support throughout the research process. The table below shows the mentor and the tasks that will be assigned to each student.

Table 2: Task assigned to each student

WP	Tasks	Assigned Students	Mentors
WP1 “Literature review”	Task 1.1 “Global and local studies on the use of turn signals and its impact on	S4 - S6	Dr. Wael & Dr. Qinaat
	Task 1.2 “Implication of turn signal timing and location”	S4 - S6	Dr. Wael & Dr. Qinaat
WP2 “Design and coding of scenarios”	Task 2.1 “Scenario design”	S1 - S6	Dr. Wael & Dr. Qinaat
	Task 2.2 “Scenario coding”	S1 - S3	Dr. Wael & Dr. Qinaat
	Task 2.3 “Questionnaire development”	S4 - S6	Dr. Wael & Dr. Qinaat

WP3 "Experiment Execution"	Task 3.1 "Participants Recruitment"	S1 – S6	Dr. Wael & Dr. Qinaat
	Task 3.2 "Data collection"	S1 - S6	Dr. Wael & Dr. Qinaat
WP4 "Data analysis and reporting"	Task 4.1 "Data extraction"	S1 - S3	Dr. Wael & Dr. Qinaat
	Task 4.2 "Data analysis"	S1 - S6	Dr. Wael & Dr. Qinaat
	Task 4.1 "Reporting"	S1 - S6	Dr. Wael & Dr. Qinaat

5. Timeline/Milestones

WPs and Tasks	Year 1											
	1	2	3	4	5	6	7	8	9	10	11	12
WP1 "Literature review"												
Task 1.1 "Global and local studies conducted on the rate of turn signals usage and its impact on road safety"												
Task 1.2 "Implication of turn signal timing and location"												
WP2 "Design and coding of scenarios"												
Task 2.1 "Scenario design"												
Task 2.2 "Scenario coding"												
Task 2.3 "Questionnaire development"												
WP3 "Experiment Execution"												
Task 3.1 "Participants Recruitment"												
Task 3.2 "Data collection"												
WP4 "Data analysis and reporting"												
Task 4.1 "Data extraction"												
Task 4.2 "Data analysis"												
Task 4.1 "Reporting"												

6. Mentoring of Students

The DS enables researchers to study the behavior and the interaction among different road users and with other roadway facilities by replicating various aspects of real-world driving conditions. It helps civil engineering students address issues related to road safety by offering a controlled and realistic environment for assessing turn signal usage and understanding the factors that influence signaling behavior. In addition, students' research skills will be improved because they will be involved in all the research phases. On the other hand, the students will work with experts from different disciplines such as traffic safety experts, civil engineers, psychologists, etc, which are all involved in the DS Laboratory at QTTSC. This will help further in improving their skills. Throughout all phases of the project, the students will receive continuous guidance and support as follows:

- ✓ Literature review: In this phase, the students will be instructed and guided on how to make a literature review. This covers a review of global and local studies conducted on the rate of turn signal usage and its timing and location implications.
- ✓ Scenario design: We will provide students with detailed examples of previously designed scenarios. This will help them in having an idea about how to design scenarios. However, we will guide and support them in designing the scenarios for this project such as road segments, buildings, other vehicles, etc.
- ✓ Scenario coding: The students will use SCANer for designing. First, a basic training session about SCANer will be delivered to the students. Moreover, we will give the students all the documents that will help them in the coding process. Students will have continuous support during this phase.
- ✓ Questionnaire development: Assisting the students on how to design clear and concise questions by selecting an appropriate response format, and structuring the questionnaire to facilitate data collection and analysis.
- ✓ Execution: We will instruct students about how to conduct driving simulator experiments. Furthermore, they will be supported in conducting a pilot study prior to the data collection process.
- ✓ Data analysis: In this phase, students will be guided on how to extract the data from the collected raw files. Furthermore, we will support them in conducting appropriate analysis tests for this project.

Reporting: Students will receive instruction on writing scientific reports, followed by ongoing feedback on their drafts.

7. Rebuttal to Peer Reviewers' Comments (for resubmitted proposals only)

We highly appreciate the valuable comments and feedback. We have addressed all reviewers' comments and made necessary adjustments to the proposal. We created an overview table to outline reviewers' comments, indicating our response and the changes made in the manuscript. We are convinced that the proposal quality is substantially improved.

Question 1: Scientific Merit**Reviewer 1**

Comment 1: The proposal did not mention using traffic safety measures from the simulator. Such data would be more compelling for the intended use of these research results: devising policy and training.

Authors Responses: Thank you for this valuable feedback. We have added a section detailing the incorporation of traffic safety measures from the simulator, including post-encroachment time (PET), time-to-collision (TTC), and braking maneuvers, to enhance the robustness of our findings.

"In addition to measuring turn signal usage, we will enhance our data collection to include traffic safety measures such as instances of hard braking, near-crashes, PET, TTC, and crashes, utilizing the simulator's capabilities to provide a comprehensive analysis of driver behavior and its impact on road safety." - Added to the "Data Collection and Analysis" section (3.4).

Comment 2: As stated in the proposal, turn signals are a "...means of communicating among vehicles and with other road users...". One issue with investigating turn signal use in a simulator is that there are no other real vehicles or road users with which to communicate. This might lead to different turn signal behavior in a simulator as compared to the real world. The proposal should have addressed how they are validating there measures with real world driving.

Authors Responses: We appreciate this insightful comment. The other vehicles in the simulator can be programmed to exhibit a range of realistic driving behaviors, including critical events such as sudden lane changes, unexpected stops, and varied signaling approaches.

"To enhance the validity of our findings, other vehicles in the simulator will be programmed to exhibit a range of realistic driving behaviors, including critical events such as sudden lane changes, unexpected stops, and varied signaling approaches." - Added to the "Scenario Design" section (3.2).

Comment 3: Finally, there is no task listed for obtaining institutional review board approval (but the proposal does mention it will be obtained). Will this be done by a student?

Authors Responses: The research mentors will handle the submission of the institutional review board (IRB) approval. However, all students involved will receive training on ethical considerations specific to driving simulator research, including safety protocols, informed consent, and data privacy.

"Obtaining the Institutional Review Board (IRB) approval for our project will be managed by the research mentors of the project." - Added to the "Procedure" section (3.3).

Reviewer 2

Comment 1: Although the authors mentioned using some statistical tools (SPSS) to analyze data, they did not mention what models would be used. I understand that it may depend on the data they collect. However, a brief discussion of potential tools would have been helpful.

Authors Responses: Thank you for your comment. We included a brief discussion of the potential statistical models that may be employed, such as regression analysis, ANOVA, and chi-square tests.

"Since every participant will be tested for each condition and situation on the driving simulator, different statistical tests by IBM SPSS 28.0 and Minitab will be used for the analyses. Potential statistical models may include regression analysis to examine the relationship between turn signal usage and socio-demographic factors, as well as ANOVA and chi-square tests to compare differences across various driving scenarios." - Added to the "Data Collection and Analysis" section (3.4).

Comment 2: Another area that lacks details is the participant selection. What is the age distribution that you will be seeking? If they select all candidates with similar age or gender, the responses could not be representative.

Authors Responses: We appreciate your feedback regarding participant selection. We will specify that our recruitment strategy aims for a diverse age distribution, targeting participants across various age groups to ensure representativeness.

"We aim to recruit a diverse participant pool through official invitation e-mails, targeting individuals aged 18 to 72 years, with a balanced representation across different age groups and genders to ensure the findings are representative of Qatar's driving population." - Added to the "Procedure" section (3.3).

Reviewer 3

Comment: From the scientific point of view, the main weakness of the proposed research is the lack of proposals to verify the results obtained in the simulator environment with the results from the real world. However, given the duration of the project, students level, verification would require additional time to complete.

Authors Responses:

We appreciate your concern regarding the verification of results. While we acknowledge the challenges posed by the project timeline and the students' experience level, we will plan for future research that includes a comparative analysis of simulator results with real-world data.

"To verify the results obtained in the simulator environment, we plan for a follow-up study that compares simulator findings with real-world driving data, acknowledging that this may require additional time and resources beyond the current project timeline." - Added to the "Data Collection and Analysis" section (3.4).

Question 2: Objective Clarity and Feasibility

Reviewer 1

Comment 1: There is a disconnect between the project title and the proposed research. The title indicates that the project will address the intent of drivers to use their signals and that the project will measure the impact of this on road safety. However, the project does not address intent (which is what the driver plans to do; that is "thinking"), but rather addresses what they do (actual use of signals).

Authors Responses: Thank you for your insightful comment. We revised the project title to better reflect the focus on actual turn signal usage rather than intent. This will clarify the scope of our research and align it with the objectives outlined in the proposal.

The title is changed from "Signaling Intent:" to "Signaling Use:"

Comment 2: Further, there are no traffic safety measures being collected. The proposed measures are about use (e.g., numbers, timing, and location). The post-survey will ask about risk perceptions on the use of signals, but this is not impact. The simulator has the ability to collect safety related information, such as hard braking, ear-crashes, and crashes so why not collect these measures?

Authors Responses: Indeed, the simulator is capable of collecting a vast number of parameters, which will be used to analyze the driving behavior in details. We added a few sentences to describe the collection of safety measures.

"In addition to measuring turn signal usage, we will enhance our data collection to include traffic safety measures such as instances of hard braking, near-crashes, PET, TTC, and crashes, utilizing the simulator's capabilities to provide a comprehensive analysis of driver behavior and its impact on road safety." - Added to the "Data Collection and Analysis" section (3.4)

Comment 3: I note that no funds have been included for participant payment.

Authors Responses: Thank you for highlighting this oversight. QRD1 does not permit participant payment to be included in the proposal budget. However, if the project is approved, we will actively seek external funding to support participant compensation.

Reviewer 2

Comment: The measurability of the achievable objective is slightly tricky and not addressed clearly in the proposal. However, with such a scope, it is difficult to address it well.

Authors Responses: Thank you for your comment regarding the measurability of our objectives. In our study, we will focus on two primary aspects:

1. Signal Use of a Virtual Car: We will investigate how the presence of a virtual car using turn signals affects the driver's reactions. Safety measurements will include parameters such as Post-encroachment Time (PET) and Time-to-Collision (TTC). This analysis will help us understand the impact of signal use on driver behavior and safety.

2. Driver's Use of Signals: We will also examine the actual use of turn signals by drivers at various locations, including roundabouts and intersections. This will provide insights into the patterns of signal usage in different traffic scenarios and its correlation with road safety.

By addressing these two areas, we aim to enhance the clarity and measurability of our research objectives, ultimately contributing to a better understanding of turn signal usage and its implications for road safety.

"In this study, we aim to investigate two key aspects of turn signal usage at different road facilities and their implications for road safety. This may include intersections (signalized, unsignalized, and roundabout), lane changes, on-ramp, off-ramp, and temporary stops. First, we will analyze the signal use of a virtual car approaching the driver. We will also examine how this influences the driver's reactions, utilizing safety measurements such as Post-encroachment Time (PET), which is the time interval between when one vehicle enters the space occupied by another vehicle and when the second vehicle reaches that same space, and Time-to-Collision (TTC), which is the estimated time remaining before a collision occurs if both vehicles continue on their current paths. Second, we will examine the actual use of turn signals by drivers at various locations, including roundabouts and intersections, to understand the patterns of signal usage in different traffic scenarios. This comprehensive approach will enhance our understanding of how turn signal behavior affects road safety and inform future policies and driver training programs." - Added to the "Objectives/Significance" section (2).

Reviewer 3

No Comments Provided

Question 3: Relevance

No Comments Provided from all reviewers

Question 4: Research Team Qualification

No Comments Provided from all reviewers

Question 5: Student Mentoring Plan

Reviewer 1

Comment: The plan does not include processes for meeting with and providing feedback to students. At a minimum it seem that there should be regularly scheduled meeting where students present what they are working on, what challenges they have faced, and how they are moved forward. This may be planned, but it was not stated explicitly in the proposal.

Authors Responses: Thank you for your valuable feedback regarding student engagement and feedback processes. Indeed, this is always in our plans, however, we missed to mention in the previous proposal. We have added information regarding it now in the proposal.

"We will implement a structured schedule of regular bi-weekly meetings where students can present their progress, discuss challenges they are facing, and receive constructive feedback from mentors and peers, ensuring continuous support throughout the research process." - Added to the "Role of the Undergraduate Researchers" section (4)

Reviewer 2

Comment: I would have liked the proposal to contain details of data analysis. Although it may depend on the data collected, a list of potential tools/models would have helped me understand the project's scope better.

Authors Responses: Thank you for your comment. We included a brief discussion of the potential statistical models that may be employed, such as regression analysis, ANOVA, and chi-square tests.

"Since every participant will be tested for each condition and situation on the driving simulator, different statistical tests by IBM SPSS 28.0 and Minitab will be used for the analyses. Potential statistical models may include regression analysis to examine the relationship between turn signal usage and socio-demographic factors, as well as ANOVA and chi-square tests to compare differences across various driving scenarios." - Added to the "Data Collection and Analysis" section (3.4)

Reviewer 3

No Comments Provided

Question 6: Overall Summary

No Comments Provided from all reviewers

8. References

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